단순선형회귀

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(o, \sigma^2)$

$\hat\beta_0 + \hat\beta_1 \bar x = \bar y, \quad \hat\beta_0 \sum x_i + \hat\beta_1 \sum x_i^2 = \sum x_i y_i$

$\hat\beta_0 = \bar y - \hat\beta_1 \bar x, \quad \hat\beta_1 = \dfrac{s_{xy}}{s_{xx}}$

$e_i = y_i - \hat y_i, \quad \sum e_i = \sum x_i e_i = 0$

$\mathrm{SST} = \sum (y_i - \bar y)^2 = \sum (y_i - \hat y_i)^2 + \sum (\hat y_i - \bar y)^2$
$= \mathrm{SSE} + \mathrm{SSR}$

$\mathrm{SST} = s_{yy}, \quad \mathrm{SSR} = \hat\beta_1^2 s_{xx}$

$R^2 = \dfrac{\mathrm{SSR}}{\mathrm{SST}} = \dfrac{\hat\beta_1^2 s_{xx}}{s_{yy}} = \dfrac{s_{xy}^2}{s_{xx} s_{yy}} = \gamma^2$

$T_0 = \dfrac{\hat\theta - \theta_0}{\sqrt{\widehat{\mathrm{Var}}(\hat\theta)}}, \quad \hat\theta \pm \mathrm{t}_{\alpha/2}(\mathrm{df}) \sqrt{\widehat{\mathrm{Var}}(\hat\theta)}$

$\widehat{\sigma^2} = \dfrac{\mathrm{SSE}}{n-2} = \mathrm{MSE}, \quad \mathrm{SSE}/\sigma^2 \sim \chi^2(n-2)$

$\mathbb{E}(\chi^2(r)) = r, \quad \mathrm{Var}(\chi^2(r)) = 2r$

$\hat\beta_1 \sim \mathcal{N}\left(\beta_1, \dfrac{\sigma^2}{s_{xx}}\right)$

$\hat\beta_0 \sim \mathcal{N}\left(\beta_0, \sigma^2 \left(\dfrac{1}{n} + \dfrac{\bar x^2}{s_{xx}}\right)\right)$

$\hat\mu_{y\cdot x} \sim \mathcal{N}\left(\beta_0 + \beta_1 x, \sigma^2 \left(\dfrac{1}{n} + \dfrac{(x-\bar x)^2}{s_{xx}}\right)\right)$

$e_i \sim \mathcal{N}\left(0, \sigma^2 \left(1 - \dfrac{1}{n} - \dfrac{(x_i - \bar x)^2}{s_{xx}}\right)\right)$

Box-Cox 변환

$$z_i^{(\lambda)} = \begin{cases} \dfrac{y_i^\lambda - 1}{\lambda} \dfrac{1}{\left(\prod y_i^{\lambda-1}\right)^{1/n}}, & \lambda \neq 0 \\ \log y_i \cdot \left(\prod y_i\right)^{1/n}, & \lambda = 0 \end{cases}$$

측정오차

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad x_i^* = x_i + \delta_i$

$\delta_i \sim \mathcal{N}(0, \sigma_\delta^2) \perp \epsilon_i, \quad \mathrm{data} : (y_i, x_i^*)$

$\mathrm{Assumption} : \dfrac{1}{n} \sum (x_i - \bar x)^2 \xrightarrow{p} \eta_x^2$

$\hat\beta_1 = \dfrac{s_{x^* y}}{s_{x^* x^*}} \xrightarrow{p} \dfrac{\eta_x^2}{\eta_x^2 + \sigma_\delta^2} \beta_1$

회귀선의 비교

$y_{1j} = \beta_{10} + \beta_{11} x_{1j} + \epsilon_{1j}, \quad \epsilon_{1j} \sim \mathcal{N}(0, \sigma^2)$

$y_{2j} = \beta_{20} + \beta_{21} x_{2j} + \epsilon_{2j}, \quad \epsilon_{2j} \sim \mathcal{N}(0, \sigma^2)$

$(\mathrm{H}_0 : \beta_{11} = \beta_{21})$

$\widehat{\sigma^2} = \dfrac{\mathrm{SSE}_1 + \mathrm{SSE}_2}{(n_1 - 2) + (n_2 - 2)}$

$\hat\beta_{11} - \hat\beta_{21} \overset{\mathrm{H}_0}{\sim} \mathcal{N}(0, \mathrm{Var}(\hat\beta_{11}) + \mathrm{Var}(\hat\beta_{21}))$

T statistics of $\hat\beta_{11} - \hat\beta_{21}$

$(\mathrm{H}_0 : \beta_{10} = \beta_{20}, \beta_{11} = \beta_{21})$

$\mathrm{SSE}_F = \mathrm{SSE}_1 + \mathrm{SSE}_2, \quad \mathrm{df}_F = n_1 + n_2 - 4$

$\mathrm{SSE}_R = \sum_i \sum_j (y_{ij} - \hat\alpha_0 - \hat\alpha_1 x_{ij})^2, \mathrm{df}_R = n_1 + n_2 - 2$

$\mathrm{F} = \dfrac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(\mathrm{df}_R - \mathrm{df}_F)}{\mathrm{SSE}_F/\mathrm{df}_F}$

$\overset{H_0}{\sim} \mathcal{F}(2, n_1 + n_2 - 4)$

이차형식

Under $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{V}), A = A^T$

1. $\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \sim \chi^2(\mathrm{r}(\boldsymbol{A}), \frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{A} \boldsymbol{\mu}) \Leftrightarrow \boldsymbol{A}\boldsymbol{V} = (\boldsymbol{A}\boldsymbol{V})^2$
2. $\mathbb{E}(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}) = \mathrm{tr}(\boldsymbol{A}\boldsymbol{V}) + \boldsymbol{\mu}^T \boldsymbol{A} \boldsymbol{\mu}$
3. $\mathrm{Cov}(\boldsymbol{y}, \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}) = \boldsymbol{V}(\boldsymbol{A} + \boldsymbol{A}^T)\boldsymbol{\mu}$
4. $\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \perp \boldsymbol{B} \boldsymbol{y} \Leftrightarrow \boldsymbol{B}\boldsymbol{V}\boldsymbol{A} = \boldsymbol{O}$
5. $\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \perp \boldsymbol{y}^T \boldsymbol{B} \boldsymbol{y} \Leftrightarrow \boldsymbol{A}\boldsymbol{V}\boldsymbol{B} \vee \boldsymbol{B}\boldsymbol{V}\boldsymbol{A} = \boldsymbol{O}$

Under $\boldsymbol{A}_j, \boldsymbol{A} = \sum \boldsymbol{A}_j$ are symmetric

$$\begin{cases} \boldsymbol{y}^T \boldsymbol{A}_j \boldsymbol{y} \sim \chi^2(r(\boldsymbol{A}_j), \frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{A} \boldsymbol{\mu}) \\ \boldsymbol{y}^T \boldsymbol{A}_j \boldsymbol{y} \text{ are independent} \\ \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \sim \chi^2(r(\boldsymbol{A}), \frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{A} \boldsymbol{\mu}) \end{cases}$$

$$\Leftrightarrow \mathrm{I} : \begin{cases} \text{Two of (a), (b), (c) hold} \\ \text{(a) } \boldsymbol{A}_j \boldsymbol{V} \text{ is idempotent} \\ \text{(b) } \forall i < j,\ \boldsymbol{A}_i \boldsymbol{V} \boldsymbol{A}_j = \boldsymbol{O} \\ \text{(c) } \boldsymbol{A}\boldsymbol{V} \text{ is idempotent} \end{cases}$$

$\mathrm{II} : \text{(c) and } \mathrm{r}(\boldsymbol{A}) = \sum \mathrm{r}(\boldsymbol{A}_j)$

Multivariate Normal Distribution

$\boldsymbol{X} \overset{d}{=} \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \boldsymbol{Z}$

$f_{\boldsymbol{X}}(\boldsymbol{x}) = |2\pi \boldsymbol{\Sigma}|^{-1/2} \exp\left(-\dfrac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\right)$

행렬의 미분

$\dfrac{\partial (\boldsymbol{c}^T \boldsymbol{x})}{\partial \boldsymbol{x}} = \boldsymbol{c}, \quad \dfrac{\partial (\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y})}{\partial \boldsymbol{y}} = (\boldsymbol{A} + \boldsymbol{A}^T)\boldsymbol{y}$

중회귀

$\boldsymbol{y}_{n\times 1} = \boldsymbol{X}_{n \times p'} \boldsymbol{\beta}_{p' \times 1} + \boldsymbol{\epsilon}_{n \times 1}$

$\boldsymbol{X}^T \boldsymbol{X} \hat{\boldsymbol{\beta}} = \boldsymbol{X}^T \boldsymbol{y}, \quad \hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y}$

$\hat{\boldsymbol{y}} = \boldsymbol{X} \hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y} = \boldsymbol{H} \boldsymbol{y}$

$\boldsymbol{H}^2 = \boldsymbol{H}, \boldsymbol{H}^T = \boldsymbol{H}, \quad \boldsymbol{H}\boldsymbol{X} = \boldsymbol{X}, \boldsymbol{H}\boldsymbol{1} = \boldsymbol{H}$

$\sum_i \boldsymbol{e}_i = \sum_i \boldsymbol{x}_{ij} \boldsymbol{e}_i = \sum_i \hat{\boldsymbol{y}}_i \boldsymbol{e}_i = 0$

$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu} = \boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n)$

$\mathrm{SST} = \boldsymbol{y}^T \left(\boldsymbol{I}_n - \dfrac{1}{n} \boldsymbol{J}_n\right) \boldsymbol{y}$

$\dfrac{\mathrm{SST}}{\sigma^2} \sim \chi^2\left(n - 1, \dfrac{1}{2} \boldsymbol{\mu}^T \left(\boldsymbol{I}_n - \dfrac{1}{n} \boldsymbol{J}_n\right) \boldsymbol{\mu}\right)$

$\mathrm{SSE} = \boldsymbol{y}^T (\boldsymbol{I}_n - \boldsymbol{H}) \boldsymbol{y}, \ \dfrac{\mathrm{SSE}}{\sigma^2} \sim \chi^2(n - p - 1)$

$\mathrm{SSR} = \boldsymbol{y}^T \left(\boldsymbol{H} - \dfrac{1}{n} \boldsymbol{J}_n\right) \boldsymbol{y}$

$\dfrac{\mathrm{SSR}}{\sigma^2} \sim \chi^2\left(p, \dfrac{1}{2} \boldsymbol{\mu}^T \left(\boldsymbol{H} - \dfrac{1}{n} \boldsymbol{J}_n\right) \boldsymbol{\mu}\right)$

$\mathrm{SSE} \perp \mathrm{SSR}, \quad \dfrac{\mathrm{MSR}}{\mathrm{MSE}} \sim \mathcal{F}(p, n - p - 1, \lambda)$

$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_n(\boldsymbol{\beta}, (\boldsymbol{X}^T \boldsymbol{X})^{-1} \sigma^2)$

$\widehat{\sigma^2} = \mathrm{MSE} = \dfrac{\mathrm{SSE}}{n - p - 1}$

변수의 직교화

$\begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \boldsymbol{O} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} = \begin{bmatrix} \cdot & \boldsymbol{O} \\ \cdot & \cdot \end{bmatrix}$

$\underset{n\times 1}{\boldsymbol{y}} = \underset{n\times(q+1)(q+1)\times 1}{\boldsymbol{X}_1 \boldsymbol{\beta}_1} + \underset{n\times(p-q)(p-q)\times 1}{\boldsymbol{X}_2 \boldsymbol{\beta}_2}$

$\begin{bmatrix} \boldsymbol{X}_2^T \boldsymbol{X}_2 & \boldsymbol{X}_2^T \boldsymbol{X}_1 \\ \boldsymbol{X}_1^T \boldsymbol{X}_2 & \boldsymbol{X}_1^T \boldsymbol{X}_1 \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}}_2 \\ \hat{\boldsymbol{\beta}}_1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{X}_2^T \boldsymbol{y} \\ \boldsymbol{X}_1^T \boldsymbol{y} \end{bmatrix}$

$\boldsymbol{X}_2^T (\boldsymbol{I}_n - \boldsymbol{H}_1) \boldsymbol{X}_2 \hat{\boldsymbol{\beta}}_2 = \boldsymbol{X}_2^T (\boldsymbol{I}_n - \boldsymbol{H}_1) \boldsymbol{y}$

$\boldsymbol{X}_2^T (\boldsymbol{I}_n - \boldsymbol{H}_1)^2 \boldsymbol{X}_2 \hat{\boldsymbol{\beta}}_2 = \boldsymbol{X}_2^T (\boldsymbol{I}_n - \boldsymbol{H}_1)^2 \boldsymbol{y}$

Let $\boldsymbol{X}_{2\cdot 1} = (\boldsymbol{I}_n - \boldsymbol{H}_1) \boldsymbol{X}_2$

$\boldsymbol{X}_{2\cdot 1}$은 $\boldsymbol{X}_2 = \boldsymbol{X}_1 \boldsymbol{\gamma}_1 + \boldsymbol{\epsilon}$ 의 잔차와 동일

$\boldsymbol{X}_{2\cdot 1}^T \boldsymbol{X}_{2\cdot 1} \hat{\boldsymbol{\beta}}_2 = \boldsymbol{X}_{2\cdot 1}^T \boldsymbol{y}$

$\hat{\boldsymbol{\beta}}_2$은 $\boldsymbol{y} = \boldsymbol{X}_{2\cdot 1} \boldsymbol{\gamma}_2 + \boldsymbol{\epsilon}$ 의 $\hat{\boldsymbol{\gamma}}_2$과 동일

$\hat{\boldsymbol{\beta}}_2$은 $\boldsymbol{e} = \boldsymbol{X}_{2\cdot 1} \boldsymbol{\gamma}_2 + \boldsymbol{\epsilon}$ 의 $\hat{\boldsymbol{\gamma}}_2$과 동일

Gauss-Markov 정리

$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y}$ is BLUE

Let $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} + \boldsymbol{A}\boldsymbol{y}$, $\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 \boldsymbol{A}\boldsymbol{A}^T$

$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}})$ positive definite

$\boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{m}$ 의 검정

$\tilde{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \mid \boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{m}\right\}$

$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})\right\}$

$\mathrm{SSE}_R = \mathrm{SSE}(\tilde{\boldsymbol{\beta}}) = \mathrm{SSE}(\hat{\boldsymbol{\beta}}) + Q = \mathrm{SSE}_F + Q$

$Q = (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{m})^T (\boldsymbol{C}(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{C}^T)^{-1} (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{m})$

$\dfrac{Q}{\sigma^2} \sim \chi^2(r(\boldsymbol{C}(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{C}^T) = k, \lambda)$

$\mathrm{SSE}(\hat{\boldsymbol{\beta}}) \sim \chi^2(n - p - 1), \quad \mathrm{SSE}(\hat{\boldsymbol{\beta}}) \perp Q$

$F_0 = \dfrac{Q/k}{\mathrm{SSE}(\hat{\boldsymbol{\beta}})/(n - p - 1)} \sim \mathcal{F}(k, n - p - 1, \lambda)$

추가제곱합

$SS(\hat\beta_0) = \boldsymbol{y}^T \dfrac{1}{n} \boldsymbol{J}_n \boldsymbol{y}$

$\hat{\boldsymbol{\beta}}_1 = (\beta_0, \dots \beta_q)^T, \hat{\boldsymbol{\beta}}_2 = (\beta_{q+1}, \dots \beta_p)^T$

$SS(\hat{\boldsymbol{\beta}}_2 | \hat{\boldsymbol{\beta}}_1) = \boldsymbol{y}^T (\boldsymbol{H} - \boldsymbol{H}_1) \boldsymbol{y}$

$SS(\hat{\boldsymbol{\beta}}_1) = \boldsymbol{y}^T \boldsymbol{H}_1 \boldsymbol{y}$

$(\boldsymbol{H} - \boldsymbol{H}_1)^2 = (\boldsymbol{H} - \boldsymbol{H}_1), \ r(\boldsymbol{H} - \boldsymbol{H}_1) = p - q$

$\dfrac{SS(\hat{\boldsymbol{\beta}}_2 | \hat{\boldsymbol{\beta}}_1)}{\sigma^2} \sim \chi^2(p - q, \lambda)$

$\lambda = SS(\hat{\boldsymbol{\beta}}_2 | \hat{\boldsymbol{\beta}}_1) \boldsymbol{\beta}_2^T \left[\boldsymbol{X}_2^T \boldsymbol{X}_2 - \boldsymbol{X}_2^T \boldsymbol{H}_1 \boldsymbol{X}_2\right] \boldsymbol{\beta}_2 / (2\sigma^2)$

$\mathrm{SSE} \perp SS(\hat{\boldsymbol{\beta}}_2 | \hat{\boldsymbol{\beta}}_1)$

Joint Confidence Region

$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x}_1 + \cdots + \beta_p \boldsymbol{x}_p + \boldsymbol{\epsilon}, \quad P(\beta \in R) \geq 1 - \alpha$

$(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T (\boldsymbol{X}^T \boldsymbol{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma^2 \sim \chi^2(p + 1), \quad \mathrm{SSE} \perp \hat{\boldsymbol{\beta}}$

$R = \left\{\boldsymbol{\beta} : \dfrac{(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T (\boldsymbol{X}^T \boldsymbol{X})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/(p + 1)}{\mathrm{SSE}/(n - p - 1)} \leq F_\alpha(p + 1, n - p - 1)\right\}$

Simultaneous Confidence Interval

$\hat{\boldsymbol{\beta}}_j \sim \mathcal{N}_n(\boldsymbol{\beta}_j, ((\boldsymbol{X}^T \boldsymbol{X})^{-1})_{(j+1)\,(j+1)} \sigma^2)$

β_j의 신뢰도가 서로 동일, 각각의 cartessian product

1. Bonferroni Method

$A_j = [\beta_j | \hat\beta_j \pm t_{\alpha/2(p+1)}(n - p - 1) \sqrt{\widehat{\mathrm{Var}}(\hat\beta_j)}]$

$P\left(\bigcap_{j=0}^p A_j\right) = 1 - P\left(\bigcup_{j=0}^p A_j^c\right) \geq 1 - \sum_{j=0}^p P(A_j)$
$= 1 - \sum_{j=0}^p \dfrac{\alpha}{p + 1} = 1 - \alpha$

2. Scheffe Method

$c^2 = (p + 1) \mathcal{F}_\alpha(p + 1, n - p - 1)$

$\forall \boldsymbol{a},\ P(\boldsymbol{a}^T \boldsymbol{\beta} \in \boldsymbol{a}^T \hat{\boldsymbol{\beta}} + c\sqrt{\boldsymbol{a}^T (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{a} \mathrm{MSE}}) \geq 1 - \alpha$

회귀모형의 비교검정

$\underset{(n_1\times 1)}{\boldsymbol{y}_1} = \underset{(n_1\times p')(p'\times 1)}{\boldsymbol{x}_1 \boldsymbol{\beta}_1} + \underset{(n_1\times 1)}{\boldsymbol{\epsilon}_1}$

$\underset{(n_2\times 1)}{\boldsymbol{y}_2} = \underset{(n_2\times p')(p'\times 1)}{\boldsymbol{x}_2 \boldsymbol{\beta}_2} + \underset{(n_2\times 1)}{\boldsymbol{\epsilon}_2}$

1. F statistics

$\boldsymbol{X} = (\boldsymbol{x}_1^T \ \boldsymbol{x}_2^T)^T, \quad \boldsymbol{y} = (\boldsymbol{y}_1^T \ \boldsymbol{y}_2^T)^T$

$\mathrm{SSE}_F = \boldsymbol{y}^T \begin{bmatrix} \boldsymbol{I} - \boldsymbol{H}_1 & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{I} - \boldsymbol{H}_2 \end{bmatrix} \boldsymbol{y}$

$\mathrm{SSE}_R = \boldsymbol{y}^T \boldsymbol{H} \boldsymbol{y}, \quad \boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T$

$\mathrm{df}_F = n - 2(p + 1), \quad \mathrm{df}_R = n - (p + 1)$

2. Hotelling's T statistics

$\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_2 \sim \mathcal{N}_{p'}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_1, ((\boldsymbol{X}_1^T \boldsymbol{X}_1)^{-1} + (\boldsymbol{X}_2^T \boldsymbol{X}_2)^{-1})\sigma^2)$

3. Using $\boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{O}$

$\boldsymbol{C}\boldsymbol{\beta} = [\boldsymbol{I}_{p'} \ -\boldsymbol{I}_{p'}] \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix} = \boldsymbol{\beta}_1 - \boldsymbol{\beta}_2$